

LIANG-JIE CHIU

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EDUCATION

Columbia University

MS in Data Science, GPA: 3.83/4.0

Expected Dec 2026

New York, NY

- Coursework: Big Data Analytics, Deep Learning, Algorithms, Time Series Forecasting, Statistical Inference & Modeling.

National Sun Yat-sen University (NSYSU)

BBA in Information Management, GPA: 3.97/4.0

Sep 2020 - Jun 2024

Kaohsiung, Taiwan

PROFESSIONAL EXPERIENCE

ASTA Health Tech

AI/ML Research Intern

Jun 2026 - Aug 2026

Jersey City, NJ (Remote)

- Engineered Gaussian (μ , σ) head for 30-minute vital-sign forecasting iTransformer, delivering calibrated 50/80/90% uncertainty bands and beating persistence by 28.5% on 1.19M-window dataset rebuilt from PostgreSQL dumps under verified patient-level splits, plus diagnostic tracking predictions going flat while vitals moved, down by half.
- Distilled ASTA's medical QA teacher traces into Qwen3-14B QLoRA student aligned to runtime chat template and JSON answer schema, lifting external MedMCQA accuracy from 0.695 to 0.739, past prior 0.720 hybrid.

TIFIN

Data Science Intern (Capstone Project)

Feb 2026 - May 2026

New York, NY (Remote)

- Designed three-layer Reasoner-Answerer-Verifier pipeline grounding financial explanations in API outputs and structured calculation descriptions — Verifier enforces 6 factual conditions and triggers corrective Reasoner retries on failure — evaluated across 227 sessions with GPT-5.4 judge on 5 metrics.
- Launched OpenAI SDK native function-calling agent orchestrating 21 APIs across 2 financial microservices, with session-aware cache enabling follow-up questions without redundant tool calls, to drive natural-language investment insights.

Research Center for Information Technology Innovation, Academia Sinica

Full-Time Research Assistant

Jul 2024 - Jul 2025

Taipei, Taiwan

Liver Tumor Segmentation with Federated Learning

- Increased liver tumor Dice score by 11% over UNet++ by applying weighted combined loss to address foreground class imbalance and HU windowing for CT preprocessing, trained on 80 GB of CT data across LiTS17 and HCC-TACE-Seg.
- Implemented Double DQN client selection framework (FAVOR) to support future federated training across distributed hospital datasets with non-IID data.

Double Spike Prediction in Phalaenopsis Orchids

- Developed two-stage Federated Learning system using YOLOv8 to extract leaf traits from orchid images and TabNet for double-spike prediction, facilitating privacy-preserving collaboration across farms with data scarcity.
- Raised small-farm model AUC from 0.69 to 0.82 by combining federated collaborative training with continual learning pipeline across federated rounds.

PROJECTS

Explainable AI for News Integrity

Fall 2025

- Architected hybrid fact-checking pipeline fine-tuning RoBERTa (F1-score 0.937) for fake news detection and building Gemini Chain-of-Thought verifier to synthesize multi-source evidence into transparent, graded verdicts.
- Integrated Perplexity Search API for real-time web verification, resolving temporal hallucinations in breaking news by cross-referencing extracted claims against live indices beyond static training cutoffs.
- Deployed serverless inference on Google Cloud Run with Cloud SQL pgvector for Wikipedia retrieval and GCS-mounted models, switching between local ChromaDB and managed services via environment-aware config.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, R, SQL, JavaScript/jQuery, HTML, CSS, PHP.

Machine Learning: PyTorch, HuggingFace Transformers, PEFT/LoRA, Scikit-learn, OpenAI SDK, statsmodels.

Data Engineering & Analytics Tools: Spark, Google Cloud Platform (GCP), BigQuery, Airflow, Hadoop, pandas, polars.